
The Jacobian lens across medical fine-tuning

Phase 1 — first real trajectory: Gemma-3-1b-it

A training-dynamics / interpretability study. We re-fit the average input→output Jacobian lens (J_ℓ) at every checkpoint and track how the internal read-out and its faithfulness to the model's own output evolve as medical SFT proceeds.

base + 14 log-spaced checkpoints · fit on 1000×128 wikitext (110,741 source positions) ·

2026-07-23

What was measured

Model / run

- Gemma-3-1b-it, full fine-tune on `mix_default`
- $\text{lr } 1 \times 10^{-5}$, 2 epochs, seed 42, 4882 steps
- 26 layers, $d_{\text{model}}=1152$
- log-spaced checkpoints $\{1, 2, 4, \dots, 4096, 4882\}$; $t=0$ is the untouched base

Per checkpoint (re-fit J_ℓ each time)

- **Drift** of J_ℓ vs the $t=0$ lens
- **Faithfulness**: $\text{KL}(\text{model} \parallel \text{lens})$ and top-1 agreement, per source layer

The lens is a function of the weights, so it is re-fit on a *frozen* generic corpus at every checkpoint.

Faithfulness is read on a *frozen* probe set split into a **general** control and a **medical** arm — the study watches medical concordance move *relative to* the general control.

Compute: ~ 9.6 h train + 14 h probe on $1 \times \text{A100}$.

The three metrics, defined

Let J_ℓ be the fitted Jacobian at source layer ℓ , J_ℓ^0 its $t=0$ (base) value, and at a probe position let p be the model's next-token distribution and q_ℓ the lens read-out transported from layer ℓ .

Drift (per layer, then mean)

$$r_\ell = \frac{\|J_\ell - J_\ell^0\|_F}{\|J_\ell^0\|_F}, \quad c_\ell = \frac{\langle J_\ell, J_\ell^0 \rangle}{\|J_\ell\| \|J_\ell^0\|}$$

How far the lens has moved from the base. r rises from 0, c falls from 1.

Faithfulness KL

$$\text{KL}(p \parallel q_\ell) = \sum_v p_v \log \frac{p_v}{q_{\ell,v}}$$

Divergence between what the model says and what the lens reads out at layer ℓ . Lower = more faithful; sensitive at small probe sets.

Top-1 agreement

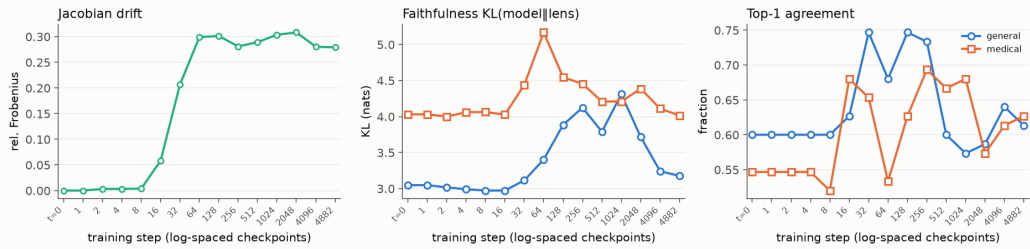
$$a_\ell = \mathbf{1}[\hat{i}_\ell = \hat{i}]$$

$\hat{i}_\ell = \arg \max q_\ell$,
 $\hat{i} = \arg \max p$: does the lens's top token match the model's?
Coarser than KL, so noisier.

What the plots show. Drift panels: $\bar{r}, \bar{c}$ meaned over all source layers. KL / top-1 panels: $\overline{\text{KL}}$ and $\bar{a}$ over the *final* $\lceil n_{\text{layers}}/5 \rceil$ layers (where the read-out has resolved), averaged over the probe prompts and split by **general** / **medical** arm. All read at the last token position.

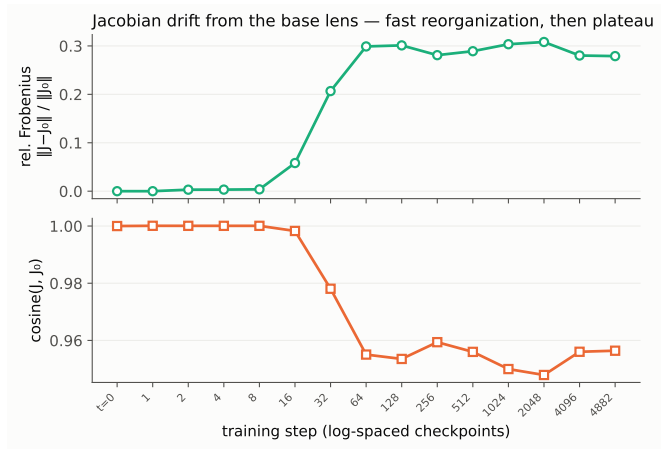
The trajectory at a glance

Gemma-3-1b-it · medical SFT · j-lens trajectory (base + 14 checkpoints)



Drift ramps then plateaus; **medical** KL sits above the **general** control throughout; top-1 agreement is noise-dominated.

Drift: fast reorganization, then plateau

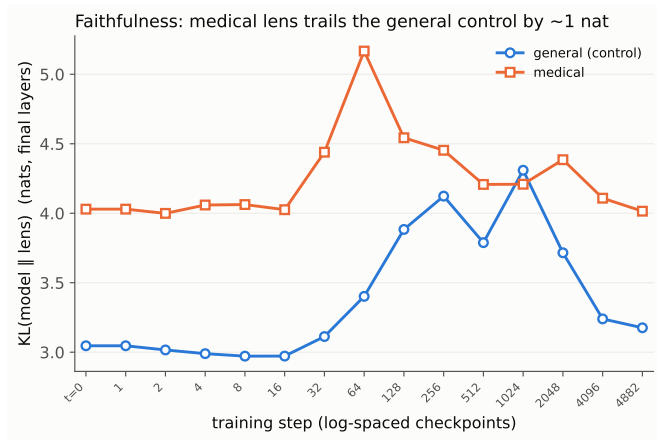


J_ℓ barely moves for the first ~ 8 steps, reorganizes sharply over steps 16–64, and **plateaus at rel. Frobenius** ≈ 0.30 (cosine ≈ 0.95) for the remaining ~ 4800 steps.

The representation's input $\rightarrow$ output coupling settles early; later training is behavioral refinement, not further lens movement.

Caveat: checkpoint-1 is bit-identical to $t=0$ (drift exactly 0) — saved before the first optimizer step landed. A wasted probe point; fix the schedule.

Faithfulness: medical trails the control by ~ 1 nat

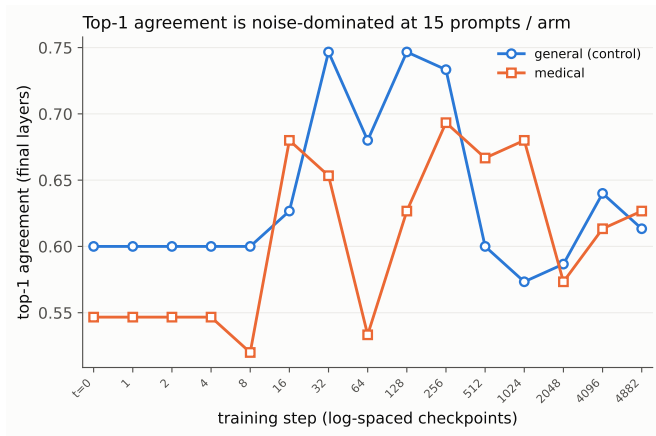


medical KL is **consistently ~ 1 nat above** the **general** control — the lens is less faithful on medical content, in the expected direction.

Both arms bump to a **mid-training peak near step 64** (**general** 3.40, **medical** 5.17) and settle back near baseline by the end (**general** 3.18, **medical** 4.01).

KL is roughly flat-with-an-excursion over the full fine-tune — not the monotone fall seen in the 270m Phase-0 toy run (different model, different mix; not expected to match).

Top-1 agreement is noise-dominated



Top-1 agreement wanders 0.52–0.75 with the two arms crossing repeatedly and **no trend** — at **15 prompts/arm** the differential signal (~ 1 nat in KL) is buried.

This is empirical support for the standing decision: **KL is the workhorse metric**; top-1 is reported but not leaned on.

Action: enlarge the probe set (medical + general cloze arms) and add jlens's 93-prompt multihop as a curated reference arm.

Findings & next steps

Findings

- J_ℓ reorganizes in the first ~ 64 steps, then plateaus — the strongest, cleanest signal in the run.
- The lens is measurably less faithful on **medical** than **general** content (~ 1 nat KL), stable across the trajectory.
- KL is smooth and readable; top-1 is noise-limited at the current probe-set size.

Fix before scaling

- checkpoint-1 $\equiv$ base — drop or shift the first log-schedule point.
- Non-resumable fit: 12b/27b need a per-step checkpoint path before a long fit can survive a requeue.

Next

- Expand + re-freeze the probe set; re-probe.
- Pair with the behavioral trajectory (MedQA / MedMCQA / PubMedQA).
- Second real trajectory: 1b-pt (the pt-vs-it starting-point axis).